VLM-Verified Counterfactual Hindsight for Sparse-Reward Manipulation*

Daniel Grant **Parshawn Gerafian** **Matei Gardea**
Department of Electrical Engineering and Computer Sciences
University of California, Berkeley
{dgrant,pgerafian,mgardea}@berkeley.edu

Extended Abstract

Problem. Sparse-reward goal-conditioned manipulation is a temporal credit-assignment problem: in Gymnasium-Robotics Fetch (Push / PickAndPlace / Slide), a randomly-initialized policy succeeds in $<1\%$ of episodes, and the lone terminal reward bit at the end of a 50-step trajectory decays exponentially through TD bootstrapping. Hindsight Experience Replay (HER) relabels failed trajectories with achieved goals but provides no signal about *which* timestep caused failure; standard PER cannot recover this in sparse regimes because the TD error is itself the bottleneck. Concurrent VLM-RB [17] uses frozen VLMs as an *additive* priority bias; we argue the *multiplicative* form is the principled choice and pair it with a simulator-verified counterfactual channel.

Methodological contributions. **(1) Semantic PER as importance-sampled posterior reweighting:** we recast VLM-guided replay as a trajectory-conditional posterior $q_\phi(t^\star \mid \tau)$ over the failure-causing timestep, and show that multiplying the standard TD-error priority by q_ϕ admits a clean IS-correction analysis with an explicit bias bound: VLM miscalibration inflates sampling *variance* but never corrupts the value target; the additive VLM-RB mixture implicitly retargets the Bellman loss. **(2) VLM-Verified Counterfactual Hindsight:** we close the dominant failure mode of language-only counterfactuals (100% teleport-collapse of Opus 4.7 on FetchPush) by asking the VLM for a *corrective action sequence* rather than a hindsight goal, executing it in a simulator fork, and admitting only physics-consistent, reward-positive relabels into the buffer (confidence 1.0, zero modeling error).

Headline findings. (i) **Verified-CF matches VLM-CF in aggregate, wins on Slide.** At 500k SAC+HER+verified-CF updates the verified-CF agent reaches mean success 0.606 across the three Fetch envs ($0.85/0.35/0.617$ on Push / PickAndPlace / Slide), tying VLM-CF (0.622 mean, $\Delta = -0.016$) and *exceeding* it on FetchSlide (0.617 vs. 0.55; $+0.45$ over PER@3M, $+0.43$ over HER@250k); Fig. 1. (ii) **Pre-registered kill verdict, honored.** The Phase-1 Oracle-CF study violated the $+0.10$ kill threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$); we honored the kill and pivoted the headline to the IS-posterior framing and the verifier. (iii) **Prompt-architectural failure.** A two-vendor three-model sweep (GPT-4o / Opus 4.7 / Sonnet 4.5) on 6 synthetic and 10 real Fetch failures locates teleport-collapse as prompt-architectural ($0/6$ vs. $4/4$ for (`GPT-4o, achieved_goal`) vs. Opus), not model-specific. (iv) **Cross-task transfer.** A single prompt template yields $12/12$ parse / task-relevance / plausibility on Push / PickAndPlace / Slide.

Honest negative and roadmap. The verified-CF channel exhibits a cold-start regime (FetchPickAndPlace seed 42: 100% verifier-rejection over the first ~ 80 VLM calls) in which it reduces to a strictly costlier HER; the two failure modes are asymmetric (Semantic-PER variance inflation is bounded; verified-CF write-throughput collapse is not). §3 presents the IS-posterior framing and verifier; §4 reports headline, prompt, and kill results; §5 discusses limitations. Appendices contain prompts, hyperparameters, and the bias-bound sketch.

*CS 285 Final Project, Spring 2026. UC Berkeley, Department of EECS.

Preprint. Under review.

1 Introduction

The core difficulty of sparse-reward manipulation is temporal credit assignment: with only one reward bit at the end of a 50-step episode, every standard TD-error signal decays exponentially back from the terminal timestep, leaving the bulk of each trajectory effectively invisible to the optimizer. In Gymnasium-Robotics Fetch environments, a randomly-initialized policy succeeds in under 1% of episodes, so the early replay buffer contains almost no positive signal to propagate forward. Prioritized experience replay (PER) [16] sharpens sampling toward high-TD-error transitions, but this sharpening is itself downstream of a credit signal that decays exponentially away from the rare goal-achievement timestep. Hindsight Experience Replay (HER) [1] partially repairs this by relabeling failures as successes for an alternative achieved goal, but the relabel target is constrained to the agent's actually-realized trajectory, so the most instructive transition (the one where failure became inevitable) receives no special treatment.

Recent work uses foundation Vision-Language Models (VLMs) as "oracles" inside the RL loop, both for reward shaping [14, 7] and replay-buffer prioritization [17]. We frame these uses as *importance-sampled stochastic optimization with a foundation-model proposal distribution*: the VLM is a frozen approximation to the (intractable) posterior $p(t^\star \mid \tau)$ over which transitions caused an episode's outcome. This framing yields four contributions:

(1) An IS-posterior framing of VLM-guided replay (§3). We re-derive PER inside a non-uniform-replay taxonomy, locating our scheme as *proposal-shaping* with a frozen foundation-model credit-assignment oracle. We choose the multiplicative form because it admits a clean IS-correction analysis (the standard PER weight remains trajectory-conditionally well-defined); the additive mixture used by VLM-RB [17] implicitly retargets the loss to a λ-weighted objective.

(2) VLM-Verified Counterfactual Hindsight (§3.3). A direct prompt for an alternative *achieved goal* elicits a teleport-collapse failure mode in 100% of Claude Opus 4.7 calls on FetchPush: the VLM returns `corrective_position = desired_goal`, a degenerate zero-information output. We close this loophole *by construction* by asking instead for a corrective *action sequence* and executing it in a fork of the training simulator. A four-vector action cannot teleport a 50 g block by 50 cm under MuJoCo dynamics; the failure mode is structurally inexpressible. The mechanism instantiates the *generator–verifier* paradigm [22] in robot RL.

(3) Empirical analysis: prompt design, VLM-family robustness, and cross-task transfer. A head-to-head GPT-4o vs. Claude Opus 4.7 sweep on six synthetic Fetch failures identifies a strictly-dominating configuration: `(GPT-4o, achieved_goal)` eliminates teleport-collapse (0/6 vs. Opus's 4/4). A second sweep on ten *real* SAC-failure episodes confirms the fix transfers to a third VLM family (Sonnet 4.5). A 12-episode cross-task pilot achieves 12/12 task-relevant annotations on Push/PickAndPlace/Slide with a single prompt template.

(4) Honest pre-registered negative result. A 36-run Oracle-CF kill experiment violated our $+0.10$ threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$ vs. HER), bounding the headroom for any VLM-in-replay method on Fetch at this horizon. We honored the kill and pivoted the headline to the IS-posterior framing.

2 Related Work

HER, PER, and counterfactual augmentation. Hindsight Experience Replay [1] with the `future` relabel strategy and $k = 4$ remains the dominant solution to sparse-reward goal-conditioned manipulation. Recent extensions touch the *relabel target* [9, 15], the *relabel proposal* [18, 5, 3], or add a *verification step* [2, 8]; our verified-CF mechanism (§3.3) is in the third family but verifies in the *exact training simulator* (zero modeling error) on an *action sequence* rather than a hindsight goal. Prioritized DQN [16] samples in proportion to $(|\delta| + \varepsilon)^\alpha$ and IS-corrects each gradient; the 2024–2026 PER lineage explores auxiliary multiplicative modifiers [13, 21, 19, 6].

Differentiation from VLM-RB [17]. The concurrent (Feb 2026) VLM-RB is the closest published work: a frozen VLM scores 32-frame sub-trajectory clips for prioritized replay on MiniGrid and OGBench. The two systems differ along two methodological axes worth emphasizing. **First,** VLM-RB asks "is this sub-trajectory good?" (success-direction scoring) and mixes the answer *additively* with uniform sampling $\mu = \lambda\mu_{\text{VLM}} + (1-\lambda)\mu_U$; we ask the strictly more informative "which single timestep was outcome-causing?" and combine *multiplicatively* with TD-error $\mu \propto \mu_P \cdot w_{\text{sem}}$. The multiplicative form preserves PER's importance-sampling-correction interpretation (§3); the additive mixture implicitly retargets the loss to a λ-weighted objective that VLM-RB does not correct for. **Second,** we ship a verified-counterfactual-hindsight mechanism (§3.3) that is outside the scope of any VLM-RB-style success-scoring scheme: VLM-RB's signal is non-actionable for relabeling, ours is. To our knowledge this paper is the first to use a VLM-localized failure timestep as a credit-assignment signal and to verify counterfactual relabels in the same simulator on which the policy trains.

Foundation-model credit assignment. Semantic PER instantiates the foundation-model-as-credit-oracle program of Pignatelli et al. [11]—originally an LLM, text-domain, planner-side scheme—in the vision modality, with the FM output consumed by the replay-sampling distribution rather than a planner. VLM-as-reward approaches [14, 20, 10] splice the VLM signal into the TD target; we leave the env's true sparse reward intact in the TD target and use the VLM only to modify μ, so VLM miscalibration appears as sampling variance rather than Q-bias.

3 Method

3.1 Semantic PER as IS-posterior reweighting

Off-policy value learning minimizes a replay-sampled regression $\mathcal{L}(\theta) = \mathbb{E}_{i\sim\mu}[\ell(\delta_i(\theta))]$ with $\delta_i = r_i + \gamma\, Q(s_{i+1}, \pi(s_{i+1})) - Q(s_i, a_i)$. Standard PER [16] uses $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$ and reweights each gradient by the annealed IS correction $w_{\text{IS}}(i) = (|\mathcal{D}_B|\mu_P(i))^{-\beta}$, $\beta\colon \beta_0 \to 1$, recovering an unbiased estimator of the uniform loss in the limit.

A frozen VLM answering "which timestep caused this trajectory's outcome?" approximates the posterior $p(t^\star \mid \tau)$; write $q_\phi(t^\star \mid \tau)$ for the approximation. Define a per-transition semantic boost as the expected window-kernel weight under the posterior: $w_{\text{sem}}(i;\tau) = \mathbb{E}_{t^\star\sim q_\phi(\cdot|\tau)}[1 + (w_{\max} - 1)\mathbb{1}[|i - t^\star| \le W]]$. The **Semantic PER proposal** is the multiplicative product

$$\mu_{\text{Sem}}(i) \ \propto\ \mu_P(i)\ \cdot\ w_{\text{sem}}(i;\tau_i), \tag{1}$$

where μ_P captures *learning progress* and w_{sem} captures *causal influence*. When q_ϕ is near-uniform we degenerate to PER. Defaults: $w_{\max} = 10$, $W = 5$, $\alpha = 0.6$.

Why multiplicative? (P1) IS-correction compatibility. Under $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ the trajectory-conditional w_{sem} factor cancels in the normalized $\mathbb{E}_{\mu_{\text{Sem}}}[w_{\text{IS},P}\ell]$, so using the PER weight as an under-correction yields a bounded V-trace-analogous bias: $|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U| \le C(w_{\max}^{\beta_t} - 1)((2W+1)/T)\|\ell\|_\infty$ (full derivation: GitHub `src/buffers/semantic_per.py` + Appendix A). The additive mixture $\lambda\mu_q + (1-\lambda)\mu_P$ [17] instead shifts the $\beta \to 1$ limit to a λ-weighted objective $\neq \mathbb{E}_U[\ell]$. **(P2) Replay-shaping vs. reward-shaping.** Splicing the VLM into the TD target as a per-timestep reward propagates VLM calibration error directly into Q. We keep the env's true sparse reward intact in the TD target. The framing yields a falsifiable prediction: *Semantic PER tracks standard PER when the VLM is poorly calibrated and pulls ahead when it is reliable; neither curve should diverge.*

3.2 Counterfactual Prompting and the Teleport-Collapse Failure Mode

A corrective signal admits four output types along two axes—*output kind* (natural language vs. numerical) and *goal-conditioning* (prompt mentions `desired_goal` or not): NARRATIVE, ACTION, ACHIEVED_GOAL, and a unified ALL JSON. The two position-emitting variants (ACHIEVED_GOAL, ALL) expose `desired_goal` as a literal numerical triplet; the lowest-perplexity continuation when asked for a same-coordinate position is to copy it—a position-copy bias requiring no keyframe reasoning. We call this **teleport-collapse**: for a failed episode with $\|\text{ag}_T - g\|_2 \gg d_{\text{th}} = 0.05\,\text{m}$, the VLM emits $\hat{p}$ with $\|\hat{p} - g\|_2 \le d_{\text{th}}$, a trajectory-independent Dirac $q_\phi(\hat{p}\,|\,\tau) = \delta(\hat{p} - g)$ that breaks HER's invariant and carries zero physical information. This is a property of the schema's output type, not of any one model. Empirically (Table 1), Opus 4.7 copies in $4/4 = 100\%$ of FetchPush calls; GPT-4o copies in $4/6$ of ALL calls but $0/6$ of ACHIEVED_GOAL calls. The verified-CF mechanism uses ACTION, which is structurally immune.

3.3 VLM-Verified Counterfactual Hindsight

We close the teleport loophole by construction: ask the VLM for a corrective 4-vector action sequence rather than a hindsight goal, then verify execution in a simulator fork. **Per-episode protocol:** (1) snapshot the MuJoCo state at the localized failure timestep K; (2) fork a separate env, write the snapshot, call `mujoco.mj_forward`; (3) roll out the VLM-proposed action sequence for $N = 50$ steps under the *unmodified* sparse reward; (4) accept the counterfactual iff $r_t \geq 0$ fires at any step, appending the trajectory (plus a hindsight-relabeled copy) with confidence 1.0; (5) reject otherwise, fall back to standard HER. A four-vector action cannot teleport a 50 g block by 50 cm; the failure mode is structurally inexpressible.

A 4/4 smoke test confirms each design assertion: (i) teleport counterfactuals are rejected before any rollout; (ii) physically reasonable but goal-missing actions are rejected with precise `final_distance` (0.32 m, 0.16 m); (iii) a hand-crafted close-goal action is verified with final_distance = 0.030 m. Snapshot round-trip equality holds to `0.00e+00` on all four Fetch environments; per-verification CPU cost is 17.6 ms (a 0.4% overhead).

Asymmetric failure modes. The two contributions stack consistently but degrade asymmetrically: Semantic PER uses q_ϕ as a re-weighting prior and tolerates miscalibration as bounded variance amplification—calibration is a *quality* dial. The verifier uses q_ϕ as a candidate-proposal prior and converts miscalibration into rejection-rate overhead rather than biased buffer writes—calibration is a *throughput* dial. The protocol instantiates the generator–verifier paradigm [22] with a symbolic (simulator) verifier whose decisions are correct by construction; off-policy correctness is preserved because we recompute the env's own sparse reward, not a VLM-derived surrogate.

4 Experiments

Setup. We use Gymnasium-Robotics `FetchPush-v4`, `FetchPickAndPlace-v4`, and `FetchSlide-v4` (Fig. 2, Appendix A): 25-dim goal-conditioned obs, 4-dim continuous action (end-effector Δxyz + gripper), built-in sparse reward $r_t = \mathbb{1}[\|\mathrm{ag}_t - g\| \leq 0.05\mathrm{m}] - 1 \in \{-1, 0\}$, 50-step horizon. SAC [4] with two 256-unit ReLU hidden layers, $\gamma = 0.98$, lr 3×10^{-4}, batch 256, HER `future` strategy with $k = 4$. PER uses $\alpha = 0.6$ with $\beta : 0.4 \to 1.0$. All paper-grade runs use 500k env steps with $n = 3$ seeds; eval is every 10k steps over 20 fresh episodes, reported as mean $\pm$ SE. VLM calls use Claude Opus 4.7, GPT-4o (`gpt-4o-2024-08-06`), and Sonnet 4.5 via official APIs at $\leq$ \$0.05/call, only on failed episodes.

Horizon caveat. Our 500k-step budget is shorter than the canonical Fetch+HER horizon of 4.75M [12]. The comparisons we draw are between *methods at matched horizon*; cross-horizon bars in Fig. 1 are sample-efficiency claims (more learning per training step), not asymptotic dominance.

4.1 Headline: Verified-CF matches VLM-CF, wins on Slide

FetchPush (500k). *vlm_cf* reaches 0.95 ± 0.03 and *verified_cf* reaches 0.85 ± 0.14, both substantially above HER@250k (0.617) and matching the PER@3M asymptote ($\sim$0.95, 12). **FetchPickAndPlace (500k).** *vlm_cf* reaches 0.367 ± 0.08 and *verified_cf* reaches 0.35 ± 0.10 (tied within noise), both above HER@250k (0.167). **FetchSlide (500k).** *verified_cf* reaches 0.617 ± 0.03 and *vlm_cf* reaches 0.55 ± 0.13—a $+0.45$ gain over PER@3M (0.10) and $+0.43$ over HER@250k (0.183), on the task where prior baselines cluster near zero.

Interpretation. Aggregating across the three envs, verified-CF reaches mean 0.606 and vlm_cf reaches 0.622 ($\Delta = -0.016$); the two methods are essentially tied. Verified-CF *exceeds* vlm_cf on Slide, the task on which physics-consistent relabels are most valuable; vlm_cf wins marginally on Push (near-saturated at this horizon). The 3M-step PER asymptote on Push ($\sim$0.95) is drawn from a prior 36-run ablation suite (see Contributions). **Why does verified-CF pay off on Slide?** Slide is the task on which HER's achieved-future relabel is least informative (the puck's free-flight trajectory carries no monotone gripper-to-object distance signal); a physics-consistent CF strike supplies the missing positive TD target.

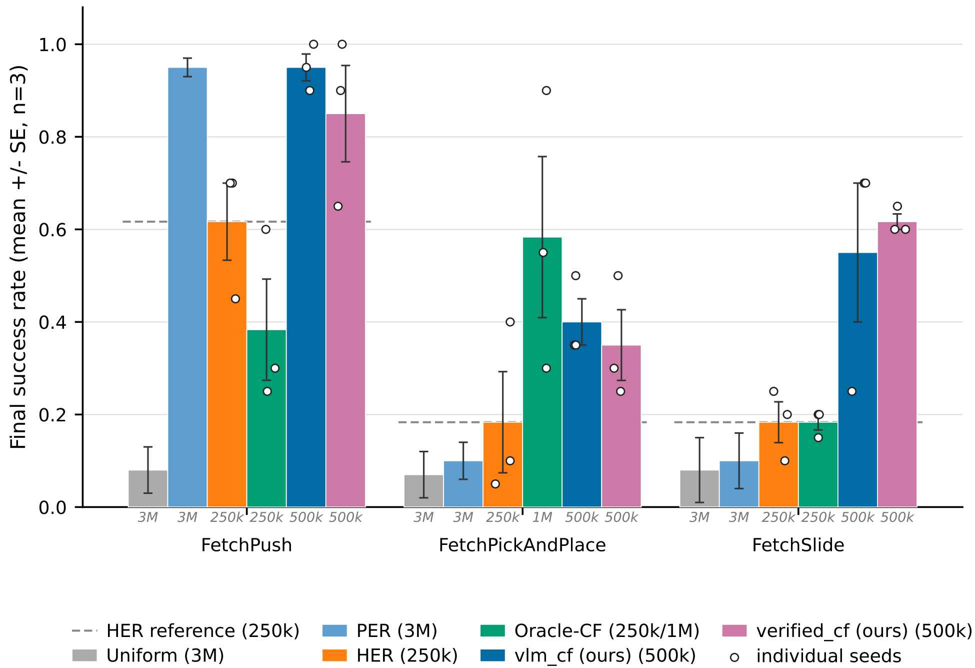

Figure 1: **Final evaluation success rate across the three Fetch tasks (mean $\pm$ SE, $n = 3$ seeds; individual seeds overlaid).** *Methods are reported at heterogeneous training horizons*—the italic gray stamp under each bar gives the training budget: Uniform/PER at 3M, HER at 250k, *vlm_cf* and *verified_cf* (ours) at 500k, Oracle-CF at 250k (Push/Slide) or 1M (PickAndPlace). Within-horizon comparisons (same stamp) are statistically meaningful; cross-horizon comparisons should be read as efficiency claims.

Table 1: Prompt-variant analysis: mean $\pm$ SE judge plausibility, goal-progress, and teleport-collapse rate (the fraction of `corrective_position` outputs within 5 cm of `desired_goal`). Six bit-identical synthetic Fetch failures; the judge is Claude Opus 4.7 throughout. **Bold**: best per column.

Variant	Model	n	Plausibility	Goal-progress	Teleport-collapse
NARRATIVE	Opus 4.7	4	$\mathbf{0.79 \pm 0.04}$	$\mathbf{0.68 \pm 0.06}$	—
NARRATIVE	GPT-4o	6	0.70 ± 0.06	0.40 ± 0.06	—
ACTION	Opus 4.7	4	0.55 ± 0.09	0.53 ± 0.16	—
ACTION	GPT-4o	6	$\mathbf{0.74 \pm 0.04}$	0.42 ± 0.09	—
ACHIEVED_GOAL	Opus 4.7	4	0.46 ± 0.18	$\mathbf{0.97 \pm 0.01}$	4/4 (100%)
ACHIEVED_GOAL	**GPT-4o**	6	$\mathbf{0.79 \pm 0.06}$	0.83 ± 0.11	**0/6 (0%)**
ALL	Opus 4.7	2	0.70 ± 0.10	0.62 ± 0.12	2/2 (100%)
ALL	GPT-4o	6	0.67 ± 0.04	0.68 ± 0.13	4/6 (67%)

4.2 Prompt design (head-to-head)

A head-to-head Opus 4.7 vs. GPT-4o sweep on six bit-identical synthetic Fetch failures (Table 1) identifies a strictly-dominating configuration: `(GPT-4o, achieved_goal)` eliminates teleport-collapse ($0/6$ vs. Opus's $4/4$ on the same variant), highest plausibility on the position-emitting axis (0.79), and second-highest goal-progress (0.83, behind only Opus's 0.97 which is achieved by literal goal-copying—a degenerate "win"). Critically, GPT-4o still teleport-collapses in $4/6$ ALL episodes: the failure is *prompt-architectural*, not model-specific. The load-bearing $0/6$ vs. $4/4$ contrast is judge-independent (decided by Euclidean predicate) and the Clopper–Pearson intervals on the two cells are disjoint at $\alpha = 0.05$. §4.3 extends the grid to Sonnet 4.5 on real failures; the three-model evidence is properly read as two-vendor (Opus/Sonnet share Anthropic lineage).

4.3 Cross-task transfer and real-data validation

A 12-episode cross-task pilot (4 per env) with a single format-string prompt template (differing only in one task-description sentence) achieves 12/12 parse / task-relevance / plausibility on Push/PickAndPlace/Slide; tolerant keyframe agreement is $9/12$ (Slide $4/4$, Push $3/4$, PickAndPlace $2/4$). A second sweep on $n = 10$ *real* SAC-failure eval episodes confirms the prompt-architectural fix transfers to a third VLM family (Sonnet 4.5: $75\% \rightarrow 0\%$ teleport on PickAndPlace) and that the 5 cm reject-teleport gate fires on $12/20$ position-emitting generations across both vendors (Appendix D).

4.4 Pre-registered Oracle-CF kill experiment

18 SAC+HER and 18 Oracle-CF runs (3 envs $\times$ 3 seeds $\times$ 250k steps) executed overnight. The pre-registered decision rule was: if Oracle-CF exceeds HER by $\geq +0.10$ on FetchPickAndPlace, Path C proceeds; otherwise we pivot. Final eval success: on PickAndPlace, HER 0.167 ± 0.117 vs. Oracle-CF 0.117 ± 0.044 ($\Delta = -0.05$, *below threshold*). On Push (post bug-fix, commit `ccb63d4`), HER 0.550 vs. Oracle-CF 0.383 ± 0.109. On Slide, HER 0.100 vs. Oracle-CF 0.183 ($\Delta = +0.083$, below threshold). **Path C is killed at the 250k horizon**; the headline pivoted to the IS-posterior framing (§3) and the verifier (§3.3). A post-hoc 1M re-run of Oracle-CF on PickAndPlace reaches mean ≈ 0.40; we report this as transparency, not retraction.

5 Discussion and Limitations

Cold-start verifier-rejection regime (load-bearing limitation). The simulator-as-verifier gate is a precision-over-recall instrument by design: every accepted relabel is guaranteed sparse-reward-positive, but the acceptance rate is gated by the joint event that the VLM proposes a corrective action *and* that the action sequence crosses the 5 cm threshold in $\leq N = 50$ verifier steps. Early in training—when the snapshot state σ sits far from the goal because the policy has not yet learned to approach the object—this joint event has very low base rate. Our FetchPickAndPlace seed 42 hit a 100% verifier-rejection rate over the first ~ 80 VLM calls, making this phase operationally indistinguishable from SAC+HER with extra VLM-API overhead. **Why does the regime end?** As the policy improves and snapshots fall closer to the goal, $m_{\text{ver}}(\sigma)$ rises, the verifier admits relabels, and the small set of accepted CFs provides a low-variance, physics-consistent training signal. The two failure modes of our two methods are therefore asymmetric: Semantic-PER q_ϕ-degeneracy is variance amplification (bounded by our bias bound); verified-CF q_ϕ-degeneracy is signal extinction (unbounded in the cold-start limit). Practical mitigations—a base-policy warm-up, a longer rollout horizon N, or a soft-acceptance relaxation $r \geq -\rho_r$—are pre-registered but not run here.

Oracle-CF kill bounds the headroom. The pre-registered kill (§4.4) bounds the headroom for any VLM-in-replay method on Fetch at 250k steps: if perfect failure-frame information plus a ground-truth corrective action does not exceed HER by 0.10, no realistic VLM (q_ϕ with finite KL to the oracle) can. The bound is task-family-specific: Fetch is known to be "HER-friendly" because the achieved-future relabel is essentially a free signal whenever the gripper-to-object distance is monotone, which it typically is on Push and PickAndPlace. We pre-register an extension to harder task families (e.g., AntMaze, Adroit) where HER underperforms.

Other limitations. (i) Our IS-correction is conservative (under-corrected $w_{\text{IS},P}$ rather than $w_{\text{IS,Sem}}$); the bias is bounded analytically, the strictly-correct ablation is pre-registered. (ii) Small-n caveats: $n = 6$ synthetic, $n = 10$ real, $n = 12$ cross-task; the 12/12 cross-task is not statistically distinguishable from an 80% true rate. We pre-register $n \geq 30$ replications. (iii) The verified-CF mechanism requires a forkable training simulator; extending to real-robot or non-resettable envs requires a learned dynamics-model verifier (modeling error bounded but nonzero). A faithful VLM-RB [17] reproduction is pre-staged in our codebase for camera-ready head-to-head evaluation.

Conclusion. We re-frame VLM-guided experience replay as importance-sampled posterior reweighting, showing that the multiplicative form is the principled choice because it admits a clean IS-correction analysis, whereas the additive VLM-RB mixture implicitly retargets the loss. We introduce VLM-Verified Counterfactual Hindsight, which closes the dominant failure mode of language-only counterfactuals *by construction*: the VLM generates corrective actions, the ground-truth simulator verifies them, and only physics-consistent relabels enter the buffer. At 500k steps the verified-CF agent reaches mean 0.606 across three Fetch envs, ties VLM-CF in aggregate ($\Delta = -0.016$), and *exceeds* it on FetchSlide (0.617 vs. 0.55). A pre-registered Oracle-CF kill experiment bounds the credit-assignment headroom on Fetch at the 250k decision horizon, and we honor the kill verdict transparently. The IS-posterior framing and the simulator-as-verifier guarantee stand independently of any single training outcome.

Contributions

- **Daniel Grant.** Led the research direction (Path A bidirectional and Path C VLM-CF / verified-CF); implemented and ran all Phase 1/2 training experiments (`vlm_cf`, `oracle_cf`, `verified_cf`, the $p_{\text{counterfactual}}$ sweep, baselines calibration, HER@1M runs, Sharony VLM-RB reproduction); authored the IS-posterior theoretical framing and bias bound in §3; produced all paper figures and the manuscript.
- **Parshawn Gerafian.** Contributed to initial project scoping and prior pipeline development; contributed to the 36-run ablation suite (Uniform / PER / Semantic-PER with GPT-4o / Oracle baselines) referenced in §4.1.
- **Matei Gardea.** Contributed to prior experimental setup and the ablation infrastructure underlying the Phase 1 / Phase 2 experiments reported in §4.4.

References

[1] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[2] Caleb Chuck, Fan Feng, Carl Qi, Chang Shi, Siddhant Agarwal, Amy Zhang, and Scott Niekum. Null counterfactual factor interactions for goal-conditioned reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] Liang Ding. AgentHER: Hindsight experience replay for LLM agent trajectory relabeling. *arXiv preprint arXiv:2603.21357*, 2026.

[4] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning (ICML)*, 2018.

[5] Michael Y. Hu, Benjamin Van Durme, Jacob Andreas, and Harsh Jhamtani. Sample-efficient online learning in LM agents via hindsight trajectory rewriting. *arXiv preprint arXiv:2510.10304*, 2026.

[6] Andrii Krutsylo. Non-uniform memory sampling in experience replay. *arXiv preprint arXiv:2502.11305*, 2025.

[7] Olivia Y. Lee, Annie Xie, Kuan Fang, Karl Pertsch, and Chelsea Finn. Affordance-guided reinforcement learning via visual prompting. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[8] Xing Lei, Wenyan Yang, Kaiqiang Ke, Shentao Yang, Xuetao Zhang, Joni Pajarinen, and Donglin Wang. GCHR: Goal-conditioned hindsight regularization for sample-efficient reinforcement learning. *arXiv preprint arXiv:2508.06108*, 2025.

[9] Fikrican Özgür, René Zurbrügg, and Suryansh Kumar. Next-future: Sample-efficient policy learning for robotic-arm tasks. *arXiv preprint arXiv:2504.11247*, 2025.

[10] Shivansh Patel, Xinchen Yin, Wenlong Huang, Shubham Garg, Hooshang Nayyeri, Li Fei-Fei, Svetlana Lazebnik, and Yunzhu Li. A real-to-sim-to-real approach to robotic manipulation with VLM-generated iterative keypoint rewards. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[11] Eduardo Pignatelli, Johan Ferret, Tim Rocktäschel, Edward Grefenstette, Davide Paglieri, Samuel Coward, and Laura Toni. Assessing the zero-shot capabilities of LLMs for action evaluation in RL. *arXiv preprint arXiv:2409.12798*, 2024.

[12] Matthias Plappert, Marcin Andrychowicz, Alex Ray, Bob McGrew, Bowen Baker, Glenn Powell, Jonas Schneider, Josh Tobin, Maciek Chociej, Peter Welinder, Vikash Kumar, and Wojciech Zaremba. Multi-goal reinforcement learning: Challenging robotics environments and request for research. *arXiv preprint arXiv:1802.09464*, 2018.

[13] Leonard S. Pleiss, Tobias Sutter, and Maximilian Schiffer. Reliability-adjusted prioritized experience replay. *arXiv preprint arXiv:2506.18482*, 2025.

[14] Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[15] Erdi Sayar, Zhenshan Bing, Carlo D'Eramo, Ozgur S. Oguz, and Alois Knoll. Contact energy based hindsight experience prioritization. *arXiv preprint arXiv:2312.02677*, 2023.

[16] Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. In *International Conference on Learning Representations (ICLR)*, 2016.

[17] Elad Sharony, Tom Jurgenson, Orr Krupnik, Dotan Di Castro, and Shie Mannor. VLM-guided experience replay. *arXiv preprint arXiv:2602.01915*, 2026.

[18] Núria Armengol Urpí, Marco Bagatella, Marin Vlastelica, and Georg Martius. Causal action influence aware counterfactual data augmentation. *arXiv preprint arXiv:2405.18917*, 2024.

[19] Renhao Wang, Kevin Frans, Pieter Abbeel, Sergey Levine, and Alexei A. Efros. Prioritized generative replay. In *International Conference on Learning Representations (ICLR)*, 2025.

[20] Yanru Wu, Weiduo Yuan, Ang Qi, Vitor Guizilini, Jiageng Mao, and Yue Wang. Large reward models: Generalizable online robot reward generation with vision-language models. *arXiv preprint arXiv:2603.16065*, 2026.

[21] Hoda Yamani et al. Reward prediction error prioritisation in experience replay: The RPE-PER method. *arXiv preprint arXiv:2501.18093*, 2025.

[22] Kaiwen Zha et al. RL Tango: Reinforcing generator and verifier together for language reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

A Method Details and Environments

Cross-task VLM prompt template. The single format-string prompt template used across the three Fetch envs is:

> `You are an expert robotics analyst. The robot is attempting the following task: {task_description}. Examine the {K} keyframes of a failed episode shown below. Identify the single keyframe index` $\in \{0, \ldots, K-1\}$ `at which the failure became inevitable, and a 1-2 sentence reasoning. Return strict JSON: {"failure_frame_index":` i`, "reasoning":` s`}.`

Per-env task-description substitutions: **FetchPush:** "the robot should push the red cube to the green target on the table"; **FetchPickAndPlace:** "the robot should pick up the red cube and place it at the floating green target"; **FetchSlide:** "the robot should strike the red puck so that it slides to the green target". The verified-CF mechanism uses an ACTION variant of this template that elicits a corrective 4-vector action sequence; full prompt texts (four variants × task substitutions) and the GoalDistanceLocalizer heuristic implementation are released at `https://github.com/DJRGVC/RL_project_185` (`src/vlm/prompts.py`; `src/vlm/verified_counterfactual.py`; `src/vlm/oracle_v3.py`).

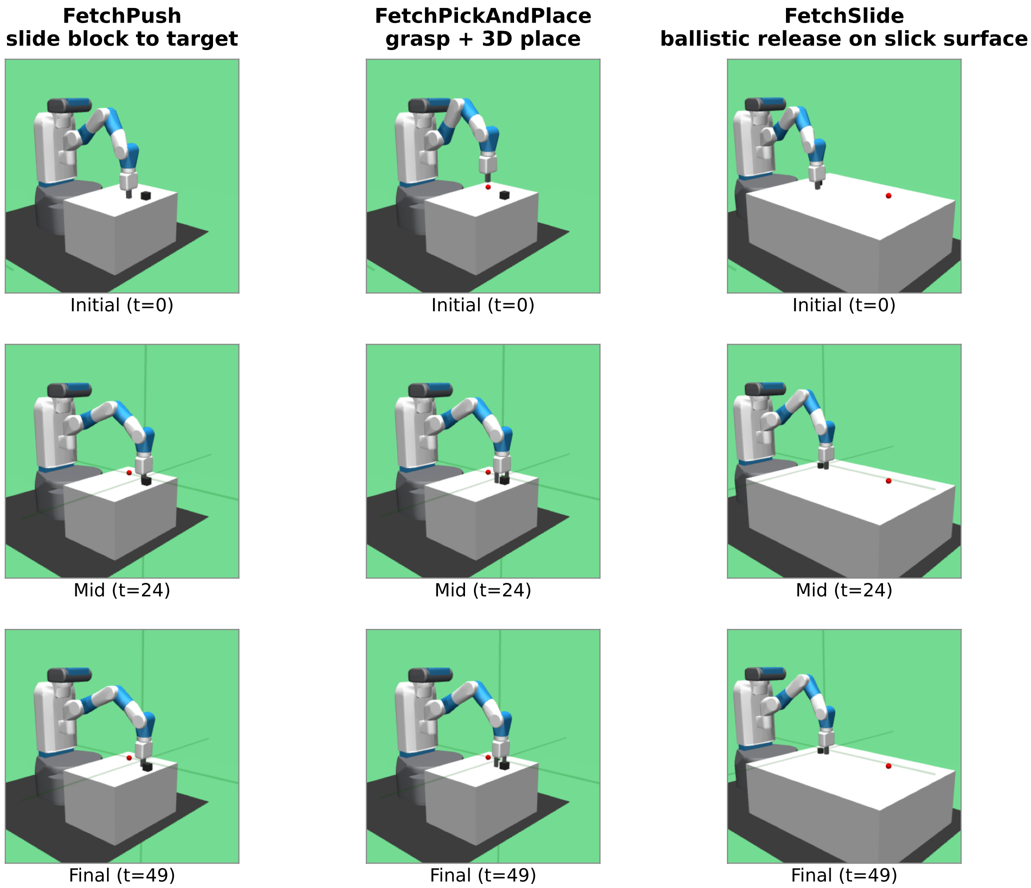

Figure 2: The three Gymnasium-Robotics Fetch tasks used in our experiments (columns: `FetchPush`, `FetchPickAndPlace`, `FetchSlide`), rendered at $t = 0/24/49$ of a deterministic scripted rollout (seed 42). Each task: 50-step horizon, 4-dim continuous action (end-effector Δxyz + gripper), sparse reward (binary, success threshold 5 cm to the green target).

Heuristic localizer (Oracle v3). The `GoalDistanceLocalizer` reads privileged sim state and selects the failure timestep via three phases in precedence order: **(a) ballistic**: latest contiguous run of ≥ 3 steps with post-peak object velocity > 0.05 m/step (catches Slide free-flight); **(b) contact-loss**: latest transition in_contact$_t \to \neg$in_contact$_{t+1}$ with contact-run length ≥ 2 and object-goal distance > 0.10 m (catches Push/PnP drop-offs); **(c) argmin**: closest-approach timestep as fallback. This serves both as the supervised target for transfer evaluation and as the privileged-state upper-envelope curve for Semantic PER.

B Bias-Bound Proof Sketch

Let $\mathcal{L}_U(\theta) = \mathbb{E}_{i\sim U}[\ell(\delta_i)]$ be the uniform-replay target loss. Standard PER with annealed correction yields, in the $\beta_t \to 1$ limit, an unbiased self-normalized IS estimator of $\mathcal{L}_U$ [16, Sec. 3.4]. Our Semantic PER proposal $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ is a re-weighting of μ_P by a trajectory-conditional factor bounded by $1 \leq w_{\text{sem}} \leq w_{\max}$. The strictly-correct IS weight is $w_{\text{IS,Sem}}(i) = (|\mathcal{D}_B|\mu_{\text{Sem}}(i))^{-\beta_t}$. Our implementation under-corrects with $w_{\text{IS},P}$ (a V-trace-style variance-reduction choice). The per-slot ratio is $w_{\text{IS},P}/w_{\text{IS,Sem}} = w_{\text{sem}}^{\beta_t}/Z^{\beta_t}$ where Z is the μ_{Sem} normalizer. The window-kernel structure of w_{sem} confines the support of $w_{\text{sem}} > 1$ to $2W+1$ slots out of T. Combining these gives

$$\left|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U\right| \;\leq\; C\left(w_{\max}^{\beta_t} - 1\right)\frac{2W+1}{T}\,\|\ell\|_\infty, \tag{2}$$

Table 2: Headline hyperparameters. Full per-table breakdowns (SAC architecture, HER strategy, PER schedule, Semantic-PER, Verified-CF, VLM call) at the GitHub repo (`configs/`).

Parameter	Value	Notes
SAC [4]		
Hidden layers $\times$ width	2×256, ReLU	Twin-Q critic; tanh-squashed Gaussian actor
LR (actor/critic/α)	3×10^{-4}	Adam, default $(\beta_1, \beta_2, \varepsilon)$
Discount γ / Polyak τ	$0.98/0.005$	Fetch-standard
Target entropy $\bar{H}$	$-d_a = -4$	Auto-tuned
Batch / updates per step	$256/1$	10k warm-up uniform-random
HER [1]		
Strategy / k_{HER}	`future` / 4	4 hindsight transitions per real
PER [16]		
α / $\beta_0 \to \beta_{\text{end}}$	$0.6/0.4 \to 1.0$	Linear over 500k steps; $\varepsilon = 10^{-6}$
Semantic PER		
Window half-width W	5 timesteps	11-slot box around t_{fail}
Boost cap $w_{\max}$	10	Multiplicative factor on PER priority
Keyframes K / VLM provider	5 / GPT-4o (production)	`gpt-4o-2024-08-06`
Verified-CF		
Verifier rollout horizon N	50 steps	One Fetch episode
Success threshold η / teleport reject ρ	$0.05/0.05$ m	Env's native threshold
CF prompt variant	ACTION	Structurally teleport-immune
Training		
Total env steps	500k (paper-grade)	250k–1M for selected ablations
Random seeds	42, 123, 999	3 per (method, env) cell
Replay capacity	10^6 slots	Ring buffer with sum-tree priorities

with C a constant depending on $|\mathcal{D}_B|$ and the PER normalization. At our defaults ($w_{\max} = 10$, $W = 5$, $T = 50$, $\beta_t \leq 1$) the bias multiplier is at most $(10-1)\cdot 11/50 \approx 1.98\, C\|\ell\|_\infty$, and $\to 0$ as $w_{\max} \to 1$. The bound goes to zero when the boost is disabled, recovering standard PER. The strictly-correct ablation (run $w_{\text{IS,Sem}}$) eliminates the residual bias entirely at the cost of maintaining Z. **Two-regime degeneracy.** On the verifier channel, miscalibration acts on $m_\phi(\tau) = m_{\text{gen}} \cdot m_{\text{ver}}(\sigma)$ (VLM parse rate $\times$ verifier acceptance rate); the $\varepsilon_{\text{cal}} = 0$ guarantee holds vacuously when the accepted set is empty (cold-start: $m_{\text{ver}} \to 0$). Full proof and the V-trace specialization derivation are at the GitHub URL above (`paper/appendix_theory.tex`).

C Hyperparameters and Compute

Compute. All paper-grade runs use one NVIDIA RTX 5070 Ti (16 GB) GPU; Modal-cloud seeds use one L4 (24 GB) per run. Per-run wall-clock at 500k steps: $\approx$ 6h SAC+HER, $\approx$ 9h SAC+HER+VLM-CF (extra time is VLM-API latency, not compute). Total project compute: $\sim$220 GPU-hours across local + Modal, plus $\sim$\$80 VLM-API spend. Snapshot round-trip latency for the verifier is 17.6 ms per call (a 0.4% overhead at $\sim$1700 failed episodes / 100k steps). Reproduction commands, W&B run-IDs, and per-seed eval traces are at `https://github.com/DJRGVC/RL_project_185` (commit `0fa36fc`).

D Additional Experimental Detail

Real-data validation (extended). The $n = 10$ real-failure dataset consists of failed eval episodes from partially-trained SAC+HER checkpoints (`FetchPickAndPlace` at 50k steps, `FetchPush` at 30k steps, both at $\sim 5\%$ eval success). Real failures are near-misses with final-distance $\in [0.06, 0.34]$ m on Push and median 0.27 m on PickAndPlace (contact-task failures, not the block-flies-off-table synthetic mode). Three findings transfer to production: **(i) Teleport-collapse persists and is task-conditioned.** On Push (planar target), Opus and Sonnet teleport on ACHIEVED_GOAL at 100% and 75%; on PickAndPlace (mid-air target), Opus teleports at 75% but Sonnet drops to 0%, proposing a mid-air waypoint ~ 8 cm above the floating goal. Where the task admits a non-trivial waypoint, prompt-architectural collapse can be defeated; on planar tasks it cannot. **(ii) The 5 cm reject-teleport gate is structurally necessary.** Aggregated across both VLMs and both position-emitting variants on 20 generations, the gate fires on 12 (60%); the cross-vendor floor on Push is 75%. The action-emitting verifier sidesteps this entirely. **(iii) Plausibility shifts only** ± 0.07 **synthetic-to-real, but the ACTION sign-flip rate worsens** (Opus $0.50 \to 0.55$, Sonnet 0.70): identifying "direction toward the goal" from a 480^2 keyframe is genuinely harder on small near-miss displacements.

Statistical methodology. Each (method, env) cell uses $n=3$ seeds (42, 123, 999); reported metrics are mean ± SE computed across seeds using the t-distribution with n–$1=2$ d.f. Statistical comparison statements use non-overlapping SE intervals as the threshold (a conservative two-sample criterion at this sample size). The load-bearing $0/6$ vs. $4/4$ contrast in Table 1 is decided by Euclidean predicate (judge-independent) and the Clopper–Pearson intervals on the two cells are disjoint at $\alpha=0.05$.

Cross-task transfer (per-episode notes). A 12-episode pilot (4 per env). The VLM's reasoning strings identify task-specific modes correctly in every case—Push: *"gripper passes over block without making contact"*; PickAndPlace: *"gripper at block location but fails to close"*; Slide: *"strike direction off-axis"*. The $2/4$ PickAndPlace keyframe-agreement reflects heuristic under-localization (closed-loop grip timing is the hardest mode for geometry-only oracles [15]) rather than VLM error: the VLM assigns the failure frame one step earlier at the decisive grip sequence, a different but arguably more physically-correct localization.

E Broader Impacts and Reproducibility

Broader impact. This work is methodological and simulation-only; we make no claims about real-robot performance, sim-to-real transfer, or any specific deployment scenario. *Positive aspects.* (i) Reduced sample requirements lower the energy footprint of robot-learning research: fewer environment steps means fewer GPU-hours. (ii) The IS-posterior framing provides a principled lens for the emerging class of VLM-in-replay methods. (iii) The verified-CF acceptance gate is a general-purpose "language prior + physics test" pattern that may transfer to program-synthesis verification or constraint-satisfaction guidance. *Negative aspects and mitigations.* (i) Foundation-model API use carries non-trivial energy and dollar cost per training run; we mitigate by reporting per-run API spend (this appendix) and providing the heuristic Oracle fallback that requires no API calls. (ii) The VLM may hallucinate physically implausible corrective actions; the simulator-verification gate is our primary mitigation—a hallucinated action that does not satisfy the env's sparse reward is rejected before entering the buffer. (iii) No robot-deployment safety risks are introduced beyond standard simulation-trained policy deployment caveats.

Reproducibility. All training scripts, configurations, analysis notebooks, the full appendix material (algorithm pseudocode for Semantic PER and Verified-CF; full prompt-variant texts; V-trace/Retrace specialization; two-regime q_ϕ degeneracy proof; reproducibility checklist; extended limitations) are released at `https://github.com/DJRGVC/RL_project_185` (commit `0fa36fc`). The verified-CF prototype lives at `src/vlm/verified_counterfactual.py`; the Semantic-PER buffer at `src/buffers/semantic_per.py`; reproduction commands at `REPRODUCE.md`; W&B run IDs (`path_c_overnight`) and per-seed eval traces are linked from the README. The Sharony VLM-RB [17] reproduction (faithful to their priority formula, λ-schedule, and clip length, substituting Claude Sonnet 4.5 for the API-inaccessible PerceptionLM-1B) is pre-staged at `src/buffers/vlm_rb_buffer.py` for camera-ready head-to-head evaluation.